

Exact No-Go Certificates for Standard Integrability Mechanisms in the Three-Dimensional Ising Layer

Exact 3D Ising Research Program
Draft accompanying the wave-17 artifact bundle

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Abstract

The three-dimensional nearest-neighbour Ising model remains unsolved. This draft consolidates exact, finite, and explicitly scoped obstructions to several standard routes to an exact solution. For a finite layer graph $\Gamma = (V, E)$ we use $A = \sum_{v \in V} X_v$ and $B = \sum_{(ij) \in E} Z_i Z_j$, with transfer representative $V = \exp(K^* A) \exp(K B)$ up to a scalar. We prove the closed Dolan–Grady defect formula, a tridiagonal no-go for representative 3D layers, the induced-claw obstruction to term-wise Majorana bilinearity, and the exact Pauli-string centralizer theorem. We then give basis-independent spectral certificates: an exact Sylvester-inertia proof for the 2×3 layer, an independent two-prime pair-product certificate for the 2×6 layer, and an all-size anisotropic theorem for every graph with a branching vertex. An arbitrary 8×8 identical- L tetrahedron/RLLL ansatz is ruled out over characteristic zero, while the full translation-invariant scalar Kac–Ward family remains unresolved. Every statement is separated from finite computation, external input, and open scope.

1 Status and scope

The result is negative but load-bearing: it excludes named mechanisms, not exact solvability by every possible representation. “Theorem” below means a proof in the stated finite or algebraic scope; “computation” means an exact finite calculation with a checked artifact; “external” means a cited result not established by this repository. None of the results below derives the thermodynamic-limit free energy, critical exponents, magnetization, or correlation functions.

For a layer graph Γ , the standard transfer factors are

$$A = \sum_{v \in V} X_v, \quad B = \sum_{(ij) \in E} Z_i Z_j,$$

and $V = \exp(K^* A) \exp(K B)$ up to a scalar. A one-dimensional layer has maximum degree at most two; a genuine three-dimensional cubic layer has a vertex of degree at least three.

2 Local algebraic obstructions

2.1 Dolan–Grady defect

The Dolan–Grady relations are

$$[A, [A, [A, B]]] = 16[A, B], \quad [B, [B, [B, A]]] = 16[B, A].$$

They generate the Onsager algebra and are the algebraic core of the original two-dimensional solution.

Theorem 2.1 (Dolan–Grady defect). *For every finite simple graph Γ , with $d(v) = \deg(v)$ and neighbourhood $N(v)$,*

$$[A, [A, [A, B]]] - 16[A, B] = 0, \quad (1)$$

$$[B, [B, [B, A]]] - 16[B, A] = R(\Gamma), \quad (2)$$

$$R(\Gamma) = 24i \sum_{v \in V} Y_v \left((d(v) - 2) \sum_{u \sim v} Z_u + 2 \sum_{\{u, u', u''\} \subset N(v)} Z_u Z_{u'} Z_{u''} \right). \quad (3)$$

Consequently $R(\Gamma) = 0$ if and only if every vertex has degree 0 or 2. Its quartic part vanishes if and only if $\Delta(\Gamma) \leq 2$. The number of four-body Pauli terms is exactly $\sum_v \binom{d(v)}{3}$, the number of induced claws in the associated frustration graph.

The proof is local. Only edges incident to the selected vertex contribute to the nested commutators; the unique Y_v prevents cancellation between different vertices. Hence a bulk square-lattice layer, and therefore the standard 3D transfer layer, fails (DG2) by an explicit quartic operator. The exact term-by-term identity was checked on sixteen graph families, including paths, cycles, stars, grids, torus layers, and a cubic graph.

2.2 Term-wise Majorana and claw obstruction

The standard Pauli generator set has frustration graph equal to the subdivision $S(\Gamma)$: X_i is adjacent precisely to $Z_i Z_j$ for incident edges, while the X terms commute with one another and all ZZ terms commute with one another.

Theorem 2.2 (Claw obstruction). *If a set of Hermitian Pauli terms is mapped injectively, after a unitary change of basis, to Majorana bilinears $i\gamma_p \gamma_q$, its frustration graph has no induced $K_{1,3}$. Therefore the standard layer generator set can be term-wise Majorana-bilinear only when $\Delta(\Gamma) \leq 2$. In particular, every square-grid layer containing a degree-three or degree-four vertex fails this representation.*

The argument is combinatorial: two bilinears anticommute exactly when their two-element index sets intersect in one element. Three pairwise commuting neighbours of one bilinear would force two of their index sets to share one element, a contradiction. This theorem is about the displayed generating set; it does not by itself exclude a radically different decomposition.

A quantitative version addresses an enlarged Gaussian realization. If the generated real Lie algebra $\mathfrak{g}$ embeds into the span of bilinears of m fermionic modes (that is, $2m$ Majorana operators), then

$$\dim \mathfrak{g} \leq \dim \mathfrak{so}(2m) = m(2m - 1), \quad m \geq \frac{1 + \sqrt{1 + 8 \dim \mathfrak{g}}}{4}.$$

This is a dimension obstruction, not a claim that no larger nonlocal embedding can ever exist.

2.3 Tridiagonal/Askey–Wilson relations

The general tridiagonal relation is tested through

$$T_0 = \beta T_1 + \gamma^* T_2 + \rho^* T_3, \quad (T_0, T_1, T_2, T_3) = ([B, B^2 A + A B^2], [B, B A B], [B, B A + A B], [B, A]).$$

It is linear in the three unknown parameters for fixed A, B .

Theorem 2.3 (Representative tridiagonal no-go). *For the boundaryless torus layers 3×3 , 3×4 , 4×4 , 3×5 , 4×5 , and 5×5 , and for the open 2×3 and 3×3 layers, the second tridiagonal relation has no solution over $\mathbb{Q}$ (hence no solution over $\mathbb{C}$). The result is invariant under every affine redefinition $A' = aA + s$, $B' = bB + t$ with $ab \neq 0$.*

For the 4×4 torus, one Pauli coefficient gives a complete one-line certificate:

$$W = Z_0 Y_5 Z_6 Z_{10}, \quad [T_0]_W = -48, \quad [T_1]_W = [T_2]_W = [T_3]_W = 0.$$

The other failing layers carry explicit one- or four-row witnesses. The witness is the same quartic $Y_v Z_{u_1} Z_{u_2} Z_{u_3}$ that appears in the Dolan–Grady defect.

2.4 Pauli-string centralizer and flux scope

Theorem 2.4 (Pauli centralizer). *Let Γ be connected and let*

$$G = \{X_i : i \in V\} \cup \{Z_i Z_j : (ij) \in E\}.$$

The Pauli strings commuting with every element of G are exactly

$$\{I, \prod_{i \in V} X_i\}.$$

The same centralizer holds for the summed pair $\{A, B\}$ within the Pauli group.

Writing a Pauli string as $(a \mid b) \in \mathbb{F}_2^{2n}$, commutation with every X_i forces $b = 0$, and commutation with every bond forces $a_i = a_j$ along every edge. Connectedness leaves only the two displayed strings. This closes a Kitaev-style family of Pauli fluxes on the same Hilbert space, but not all possible non-Pauli symmetries, enlarged-space constructions, or ordinary lattice translation sectors.

3 Basis-independent spectral no-go results

Algebraic failure of the obvious generators does not exclude “free fermions in disguise.” The following tests use only the spectrum.

3.1 Gaussian subset-product criterion

If a positive-spectrum Gaussian operator on m modes has single-particle factors $u_1, \dots, u_m \geq 1$, its spectrum is

$$\{\lambda_0 \prod_{k \in S} u_k : S \subseteq \{1, \dots, m\}\}.$$

If the three lowest eigenvalues are distinct, then $\lambda_1 \lambda_2 / \lambda_0$ must also be an eigenvalue. For rational $t = \tanh(K^*/2)$, the symmetrized transfer representative is, up to a positive scalar,

$$R = P_t \operatorname{diag}(q^{(b_k - \epsilon)/2}) P_t, \quad P_t(k, l) = t^{\operatorname{Ham}(k, l)}, \quad q = \frac{1 + t^2}{2t},$$

which is exact rational symmetric positive definite.

Theorem 3.1 (Theorem S: exact 2×3 spectral no-go). *At $t = 1/3$ (equivalently $q = \exp(2K) = 5/3$), the open 2×3 layer transfer operator is not a fermionic Gaussian operator on six modes in any Majorana basis. Exact rational inertia counts at the two outward-rounded endpoints of the forced-eigenvalue interval are both 8, so the interval contains no eigenvalue, while a six-mode Gaussian spectrum would contain $\lambda_1 \lambda_2 / \lambda_0$.*

The control chains of lengths four and six show the opposite inertia pattern, as expected for the 2D free-fermion model. Equal inertia counts prove absence; unequal counts only show that some eigenvalue lies in the interval and are not a proof of exact presence.

3.2 The 2×6 parity-half certificate

For a 12-mode Gaussian, any subset of one parity half has at most

$$A_{\text{same}}(12) = \sum_{\substack{2 \leq r \leq 12 \\ r \text{ even}}} \binom{12}{r} 2^{12-r} = 261625$$

distinct within-half pair products. The physical ($P = +1, \rho = +1$) block of the open 2×6 layer has dimension 1056 and $\binom{1056}{2} = 557040$ pair slots.

Theorem 3.2 (Theorem 2×6 -NG). *At $t = 1/3$, the physical $P = +1$ sector of the open 2×6 layer is not either parity half of any 12-mode Gaussian subset-product multiset. On both primes 1000003 and 2000003, the exact pair-polynomial computation gives*

$$\deg \gcd(C, C') \leq 295414 < 295415 = \binom{1056}{2} - A_{\text{same}}(12).$$

Thus the block has at least 261626 distinct within-half pair products, at least one above the Gaussian ceiling.

This is a one-sided modular lower-bound argument over $\mathbb{Q}$, not a claim that the exact number of distinct products is 261626. The saved Euclidean tail was independently replayed with bit-identical remainders; the standalone verifier performs 38 checks, including artifact-key and digest-chain checks.

3.3 All-size anisotropic theorem

Theorem 3.3 (Theorem F: all-size anisotropic no-go). *Let G be any finite simple graph with a vertex of degree at least three and $n = |V| \geq 4$. For the inhomogeneous rational layer family with independent site parameters t_v and bond parameters w_e , there is a nonzero polynomial ρ_G with*

$$\deg \rho_G \leq 4(2n + |E|)S(S-1), \quad S = \binom{2^n}{2},$$

such that whenever $\rho_G \neq 0$, the eigenvalues have at least

$$3^n - 2^n + 48 \cdot 3^{n-4}$$

distinct unordered slot-pair products. A full n -mode Gaussian multiset has at most $3^n - 2^n$, so the generic inhomogeneous layer is not Gaussian.

The exceptional set is a proper algebraic hypersurface. This theorem is genuinely anisotropic: it does not cover the physical isotropic curve, where the present invariant reaches the Gaussian ceiling on the tested claw core. The named isotropic 2×3 , 2×4 , and 3×3 points are covered separately by finite or coupling-generic certificates; no isotropic all-size theorem is claimed.

4 Tensor and Kac–Ward ansatzes

4.1 Identical- L tetrahedron/RLLL equation

Let $L(q)$ be the verified six-leg Ising tensor and consider

$$R_{123}L_{145}(q)L_{246}(q)L_{356}(q) = L_{356}(q)L_{246}(q)L_{145}(q)R_{123},$$

with an arbitrary 8×8 matrix R and no auxiliary grading assumption.

Theorem 4.1 (Ungraded tetrahedron no-go). *Over every characteristic-zero field, if $q \notin \{0, 1, -1\}$, the only arbitrary 8×8 solution of the identical- L equation is $R = 0$. Therefore no invertible auxiliary R exists for finite ferromagnetic weights $0 < q < 1$.*

The proof concatenates two complementary 64×64 minors whose common factor is

$$q^{109}(q-1)^{92}(q+1)^{77};$$

the remaining factors are coprime by an explicit Bezout identity. The scope is limited to identical local factors, this vertex-type equation, and auxiliary dimension eight.

4.2 Kac–Ward status

A full translation-invariant scalar Kac–Ward transition has 30 directed-direction weights, reduced to 25 variables on a generic five-edge gauge chart. The exact 2D planar identity is reproduced as a control.

Theorem 4.2 (Positive-real scalar Kac–Ward no-go). *No assignment of nonnegative real values to all 30 allowed direction transitions satisfies the coefficientwise scalar Kac–Ward identity on every free cubic box. In fact, the order-four trace equation on the free $2 \times 2 \times 2$ box is*

$$F_4 = 48 + 8(M_{xy,+} + M_{xy,-} + M_{xz,+} + M_{xz,-} + M_{yz,+} + M_{yz,-}),$$

which is strictly positive on the nonnegative orthant. The stored rational Positivstellensatz identity certifies infeasibility on all 32 selected gauge strata.

The full complex scalar family remains unresolved. Exact order-4, 6, 8 systems have finite-field points and modular Groebner attempts timed out; a numerical construction point fails an independent $3 \times 3 \times 3$ order-eight holdout, but that does not prove ideal emptiness. Matrix-valued weights, position-dependent weights, spin-structure sums, and higher auxiliary dimensions remain outside this paper’s no-go claims.

5 Conclusion

The results jointly exclude the standard Onsager/Dolan–Grady route on branching layers; the full tridiagonal relation family for representative 3D layers; term-wise Majorana bilinearity of the standard generators; same-space Pauli-string flux decompositions; full-layer Gaussian spectra at the certified named points and generically in the anisotropic family; the displayed identical- L eight-dimensional tetrahedron ansatz; and the nonnegative-real scalar Kac–Ward corner.

They do not exclude an exact solution based on a larger Gaussian restriction, non-Pauli conserved quantities, a different matrix-valued or dynamical tensor ansatz, a nonlocal transformation, or an as-yet unknown mechanism. The full 3D Ising problem is therefore still open.

Artifact and reproduction map

The accompanying bundle contains the exact source notes, scripts, tests, and JSON artifacts. Principal paths are:

- `proofs/dolan_grady_defect.md`, `results/dg_theorem_check.json`, and `experiments/e16_dg_theorem_check.py`;
- `proofs/tridiagonal_nogo.md`, `results/tridiagonal_certificate_general.json`;
- `proofs/algebraic_obstruction.md`, `results/flux_sectors.json`;
- `proofs/spectral_gaussianity.md`, `results/spectral/gaussianity_certificate.json`;
- `proofs/gaussian_2x6.md`, `results/gaussian_2x6.json`, `results/gaussian_2x6_work/`;
- `proofs/allsize_gaussian.md`, `results/spectral/allsize_gaussian.json`;
- `proofs/tetra_ungraded.md`, `results/integrability/tetra_ungraded.json`;
- `proofs/kac_ward_positive_real.md`, `proofs/kac_ward_full_family.md`, and `results/kac_ward/`.

Run the standalone tests from the repository root with `.venv/bin/python`; the bundle's `README.md` records the acceptance gate and exact environment assumptions.